

HDOC Day 18

Name- Aaryan Pratap Singh

Sap id - 590032842

Batch- 15

1. Algorithm

Main Program

1. Start.
2. Display the menu:
 - 1 → Strong Number
 - 2 → Armstrong Number
 - 3 → Harshad Number
 - 4 → Duck Number
3. Read the user's choice.
4. Use **switch-case**.
5. Call the corresponding **void** function using a simple function call.
6. If the choice is invalid, display "Invalid choice".
7. Stop.

Function 1: Strong Number

A number is **Strong** if the sum of the factorials of its digits equals the number.

Example: $145 = 1! + 4! + 5! = 145$

1. Read a positive integer **n**.
2. Copy **n** into **temp**.
3. Initialize **sum = 0**.
4. Extract each digit using $\% 10$.
5. Calculate the factorial of the digit.
6. Add the factorial to **sum**.
7. Remove the last digit using $/ 10$.
8. Repeat until all digits are processed.
9. If **sum == n**, display that it is a Strong number; otherwise, display that it is not.

Function 2: Armstrong Number

An Armstrong number is equal to the sum of its digits raised to the power of the total number of digits.

Example: $153 = 1^3 + 5^3 + 3^3 = 153$

1. Read `n`.
2. Count the number of digits.
3. Extract each digit.
4. Calculate `digit ^ number_of_digits`.
5. Add the result to `sum`.
6. Repeat for all digits.
7. If `sum == n`, it is an Armstrong number; otherwise, it is not.

Function 3: Harshad Number

A **Harshad number** is divisible by the sum of its digits.

Example: $18 \rightarrow 1 + 8 = 9$, and $18 \% 9 = 0$.

1. Read `n`.
2. Find the sum of its digits.
3. Check `n \% sum`.
4. If the remainder is 0, it is a Harshad number.
5. Otherwise, it is not.

Function 4: Duck Number

A **Duck number** is a positive integer containing at least one 0 in its digits, excluding leading zeros.

Example: 1023 is a Duck number.

1. Read `n`.
2. Extract its digits one by one.
3. Check whether any digit equals 0.
4. If a zero is found, set `isDuck = 1`.
5. If `isDuck == 1`, display that it is a Duck number.
6. Otherwise, display that it is not.

2. Test Case Table

Test Case	Choice	Input	Expected Output	Actual Output	Result
1	1	145	Strong number	145 is a Strong number.	Pass
2	1	123	Not a Strong number	123 is not a Strong number.	Pass
3	2	153	Armstrong number	153 is an Armstrong number.	Pass
4	2	123	Not an Armstrong number	123 is not an Armstrong number.	Pass
5	3	18	Harshad number	18 is a Harshad number.	Pass
6	3	19	Not a Harshad number	19 is not a Harshad number.	Pass
7	4	1023	Duck number	1023 is a Duck number.	Pass
8	4	1234	Not a Duck number	1234 is not a Duck number.	Pass

3. C Program

```
#include <stdio.h>
/* Function declarations */
void strongNumber();
void armstrongNumber();
void harshadNumber();
void duckNumber();
int main()
{
    int choice;
    printf("1. Check Strong Number\n");
    printf("2. Check Armstrong Number\n");
    printf("3. Check Harshad Number\n");
    printf("4. Check Duck Number\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            strongNumber();
            break;
        case 2:
            armstrongNumber();
            break;
        case 3:
            harshadNumber();
            break;
        case 4:
            duckNumber();
            break;
        default:
            printf("Invalid choice\n");
    }
    return 0;
}

/* Function to check Strong Number */
void strongNumber()
{
    int n, temp, digit;
    int sum = 0, fact, i;
```

```

printf("Enter a positive integer: ");
scanf("%d", &n);
temp = n;
while (temp > 0)
{
    digit = temp % 10;

    fact = 1;

    for (i = 1; i <= digit; i++)
    {
        fact = fact * i;
    }

    sum = sum + fact;
    temp = temp / 10;
}

if (sum == n)
    printf("%d is a Strong number.\n", n);
else
    printf("%d is not a Strong number.\n", n);
}
/* Function to check Armstrong Number */
void armstrongNumber()
{
    int n, temp, digit;
    int digits = 0, sum = 0;
    int i, power;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    temp = n;
    /* Count the digits */
    while (temp > 0)
    {
        digits++;
        temp = temp / 10;
    }
    temp = n;
    /* Calculate Armstrong sum */
    while (temp > 0)
    {
        digit = temp % 10;
        power = 1;
        for (i = 1; i <= digits; i++)

```

```

        {
            power = power * digit;
        }
        sum = sum + power;
        temp = temp / 10;
    }
    if (sum == n)
        printf("%d is an Armstrong number.\n", n);
    else
        printf("%d is not an Armstrong number.\n", n);
}
/* Function to check Harshad Number */
void harshadNumber()
{
    int n, temp, digit;
    int sum = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    temp = n;
    while (temp > 0)
    {
        digit = temp % 10;
        sum = sum + digit;
        temp = temp / 10;
    }
    if (n % sum == 0)
        printf("%d is a Harshad number.\n", n);
    else
        printf("%d is not a Harshad number.\n", n);
}
/* Function to check Duck Number */
void duckNumber()
{
    int n, temp, digit;
    int isDuck = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    temp = n;
    while (temp > 0)
    {
        digit = temp % 10;
        if (digit == 0)
        {
            isDuck = 1;
            break;
        }
    }
}

```

```

    }
    temp = temp / 10;
}
if (isDuck == 1)
    printf("%d is a Duck number.\n", n);
else
    printf("%d is not a Duck number.\n", n);
}

```

4. Compilation

```

[aaryanrajput@aaryans-macbook ~ % nano day18.c
[aaryanrajput@aaryans-macbook ~ % clang day18.c -o day18.c
aaryanrajput@aaryans-macbook ~ % █

```

5. Sample Execution

Strong Number

```

[aaryanrajput@aaryans-macbook ~ % nano day18.c
[aaryanrajput@aaryans-macbook ~ % clang day18.c -o day18.c
[aaryanrajput@aaryans-macbook ~ % ./day18.c
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
Enter your choice: 1
Enter a positive integer: 145
145 is a Strong number.
aaryanrajput@aaryans-macbook ~ % █

```

Armstrong Number

```

[aaryanrajput@aaryans-macbook ~ % ./day18.c
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
Enter your choice: 2
Enter a positive integer: 153
153 is an Armstrong number.
aaryanrajput@aaryans-macbook ~ % █

```

Harshad Number

```
[aaryanrajput@aaryans-macbook ~ % ./day18.c
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
Enter your choice: 3
Enter a positive integer: 18
18 is a Harshad number.
aaryanrajput@aaryans-macbook ~ %
```

Duck Number

```
[aaryanrajput@aaryans-macbook ~ % ./day18.c
1. Check Strong Number
2. Check Armstrong Number
3. Check Harshad Number
4. Check Duck Number
Enter your choice: 4
Enter a positive integer: 1023
1023 is a Duck number.
aaryanrajput@aaryans-macbook ~ %
```

6. Testing Result

I compiled and executed the program against **all 8 prepared test cases**. All produced the expected results: